

A LIFE SAVED A military helicopter rescues a boy in a floodhit settlement on the Nepal-Tibet border, Aug. 26

have ranked Nepal as among the most vulnerable to the disastrous consequences of climate change. In a July 2026 report, Nepal's Department of Water Resources and Irrigation estimated that 4,271 people lost their lives between 2011 and 2025 to "water-induced disasters" exacerbated by rising temperatures and increasingly unpredictable weather patterns. In Nepal's mountains, glacial lake outburst floods (GLOF), attributed to the gradual thawing of permafrost caused by rising global temperatures, have become more recurrent since 1990. One portent was the June 2025 flood, also along the Bhote Koshi river, that killed nine and left 20 missing in Rasuwa. As it turned out, this incident foreshadowed a much larger catastrophe along and beyond the same route.

Ironically, Nepal remains one of the lowest contributors to global carbon emissions—less than 0.1 per cent of the world's share. This tragedy puts the nation at the centre of a global debate—the unwitting victim of a crisis it did very little to create. This was noted by PM Shah, who in his September 2 parliament address called the floods "a critical signal of the increasing risks [Nepal's] Himalayan region is facing due to climate change" and said that "it is the shared responsibility of the global community to mitigate and manage its effects".

The address followed calls for the prime minister to represent Nepal at the United Nations General Assembly and make an appeal for relief and compensation, as well as better policymaking and action around climate change globally. In the absence of an immediate and radical mass restructuring of the world's relationship to the environment, the future looks bleak. ■

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tinuously updated, catchment-level Himalayan hazard atlas. It should combine lake growth, bathymetry, glacier movement, hanging-ice geometry, slope deformation, permafrost, landslide history and downstream exposure. Radar and optical satellites can identify change, targeted field expeditions and drones must measure the most dangerous sites. The risk ranking should be public enough for communities, project developers and insurers to act on it.

As important is to have genuinely end-to-end warning systems in the most exposed basins, with redundancy from the headwaters to the plains. Every system must be stress-tested for power loss, broken towers and severed roads. Local volunteers, the Indo-Tibetan Border Police, police, dam operators and district officials need rehearsed roles, marked escape routes and regular night-time as well as daytime drills. Alongside, central and state governments should make climate-adjusted multi-hazard assessment legally binding for every dam, highway, tunnel, hotel cluster and township in the Himalaya. Existing projects must be reassessed against the probable maximum GLOF and rock-ice avalanche, including cascading dam failure. Where risk cannot be engineered down to an acceptable level, the project must be stopped, redesigned or relocated.



"EWS installation requires physical expeditions dangerous to execute in difficult high-altitude terrain that can be accessed only in narrow weather windows. And simply installing EWS is not enough, they need to be able to communicate"

KRISHNA VATSA, *Head of Department, NDMA*

Another must is to enforce land-use zoning. Hazard maps must determine where construction is prohibited, where only low-occupancy uses are allowed and where structures require flood-resistant design and vertical evacuation. Rivers need room to carry exceptional flows. The costs of relocation and retrofitting will be politically difficult, but far smaller than rebuilding a valley after it has been erased. There should also be facilities to lower the water level of the limited number of accessible lakes where engineering analysis shows that siphoning or a controlled outlet will substantially cut risk.

There should also be a compact with neighbouring countries for real-time alerts, joint scientific teams and basin-wide emergency protocols. India cannot monitor a transboundary river only after it crosses the border. Nepal's black torrent was a warning delivered at 180 kmph. It is in India and our neighbours' hands to decide whether thousands of people, a dam, a highway and an entire town are ready to take evasive measures below when it does. ■